

EX NO:1	Butterworth IIR Filters
DATE:	1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

Code:

```

clc;
clear;
close all;

% EX NO: 1
% Butterworth IIR Filters
% 1.2 Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB

% Specifications
Fs = 10000;    % Sampling Frequency (Hz)
Fp = 3000;    % Passband Frequency (Hz)
Fst = 2000;    % Stopband Frequency (Hz)

% Calculate Minimum Filter Order
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), 1, 60);

% Design Butterworth High-Pass Filter
[b, a] = butter(N, Wn, 'high');

% Magnitude & Phase Response
figure;
freqz(b, a);
title('Frequency Response');

% Impulse Response
figure;
[h, n] = impz(b, a, 50);
stem(n, h);
grid on;
title('Impulse Response');
xlabel('Samples');
ylabel('Amplitude');

% Step Response
figure;
stepz(b, a);
title('Step Response');

% Group Delay
figure;
grpdelay(b, a);
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot');

% Display Results
fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);

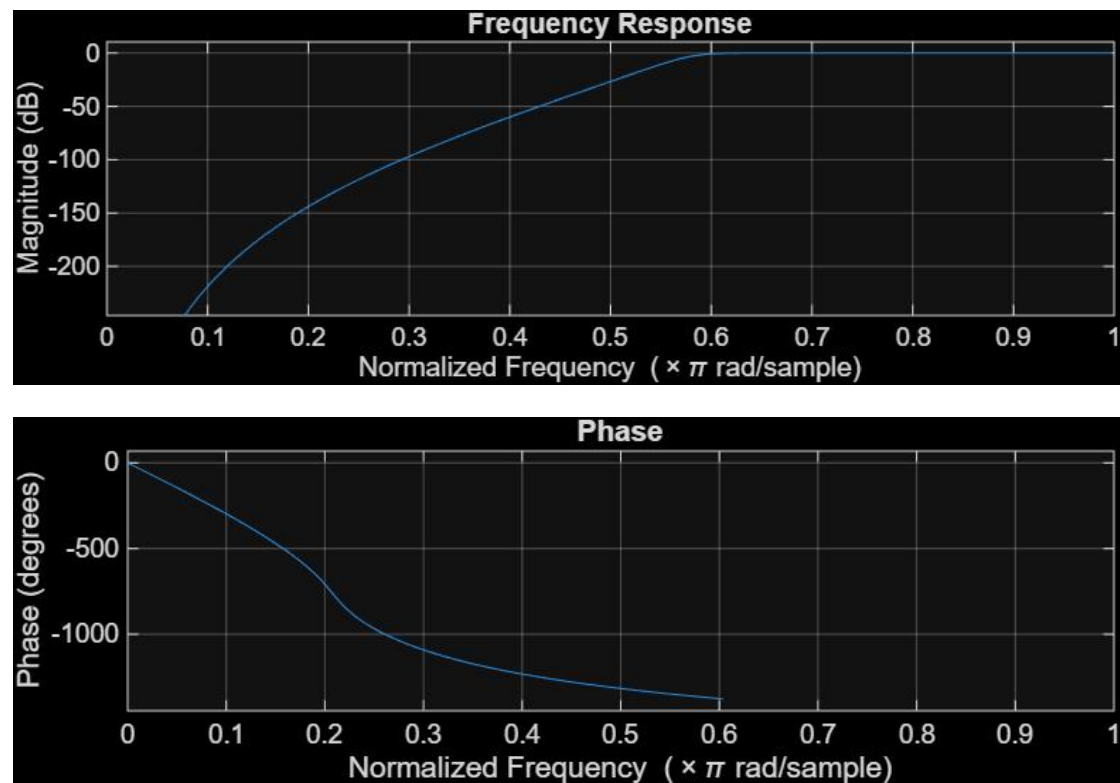
```

Outputs:

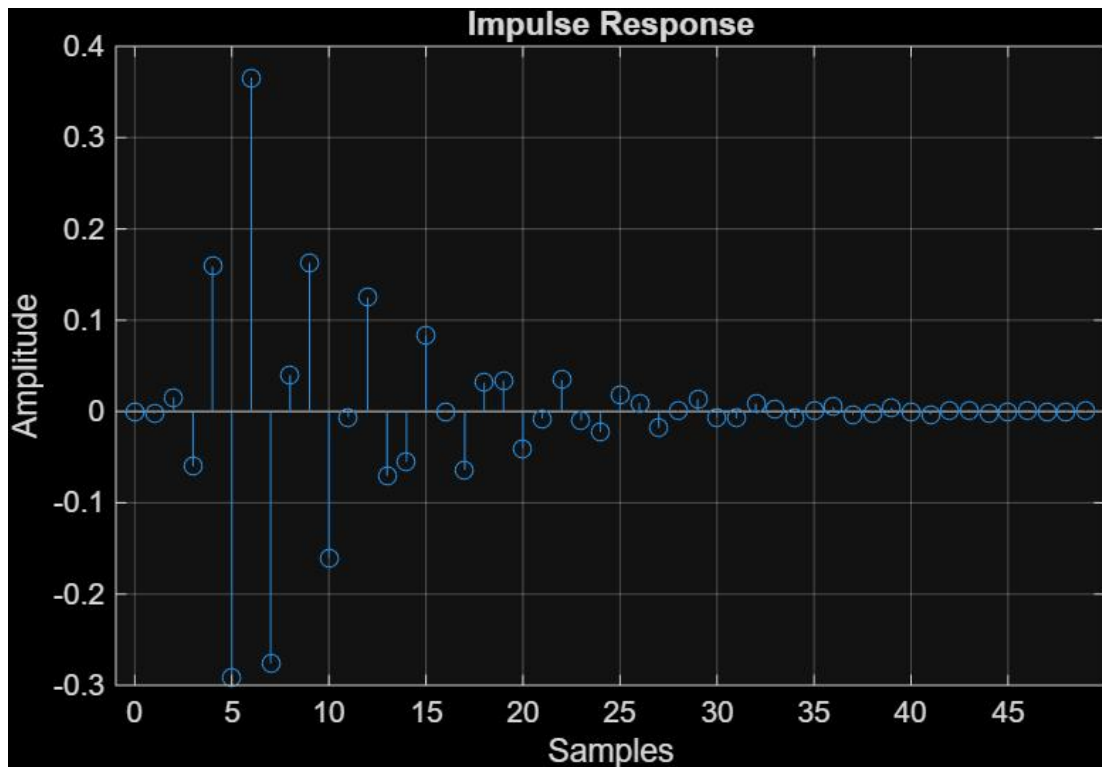
Objective 1:

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Command Window
Minimum Filter Order (N) = 12
Cutoff Frequency (Wn) = 0.5807
>>
```

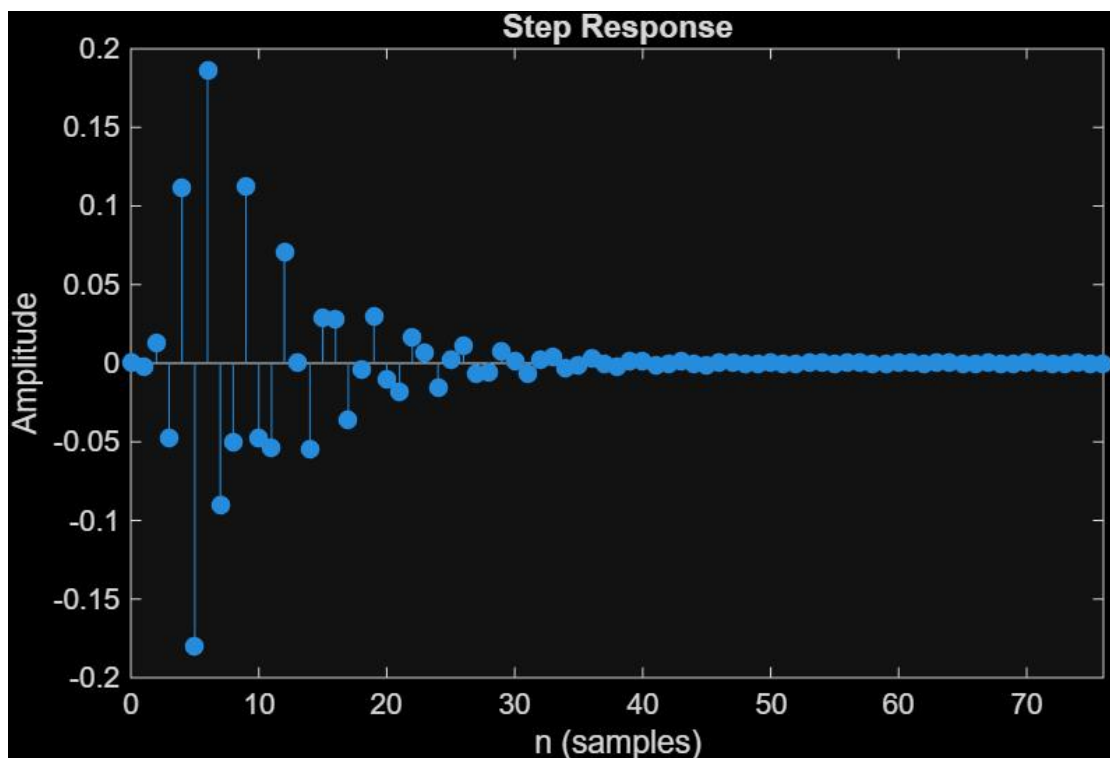
Objective 2:



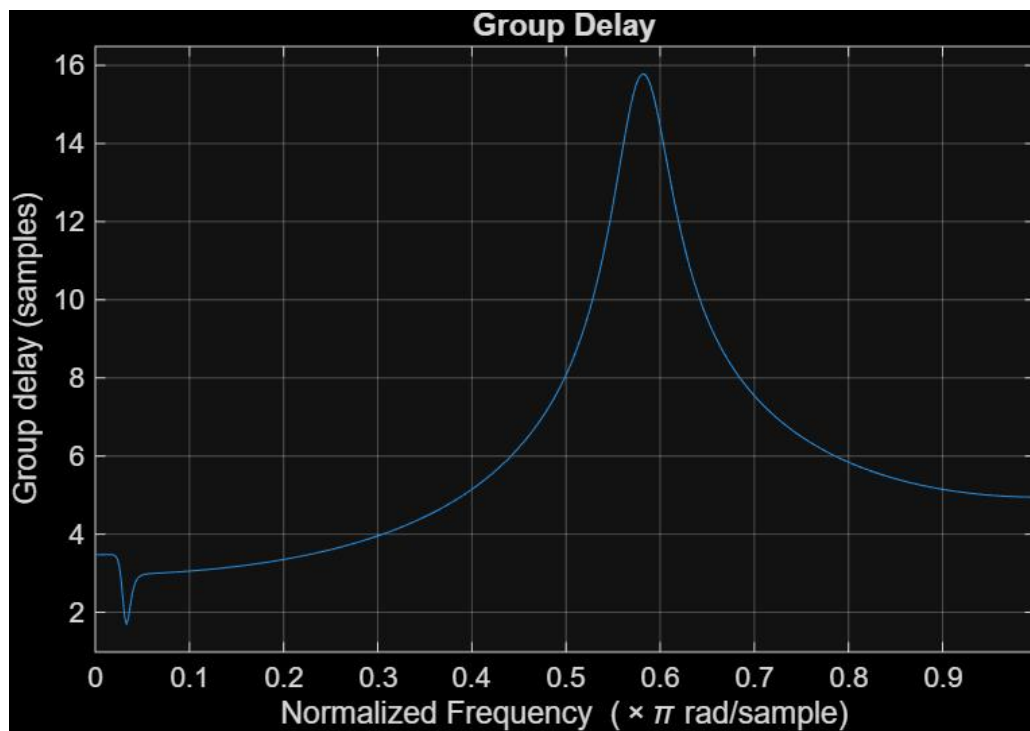
Objective 3:



Objective 4:



Objective 5:



Objective 6:

